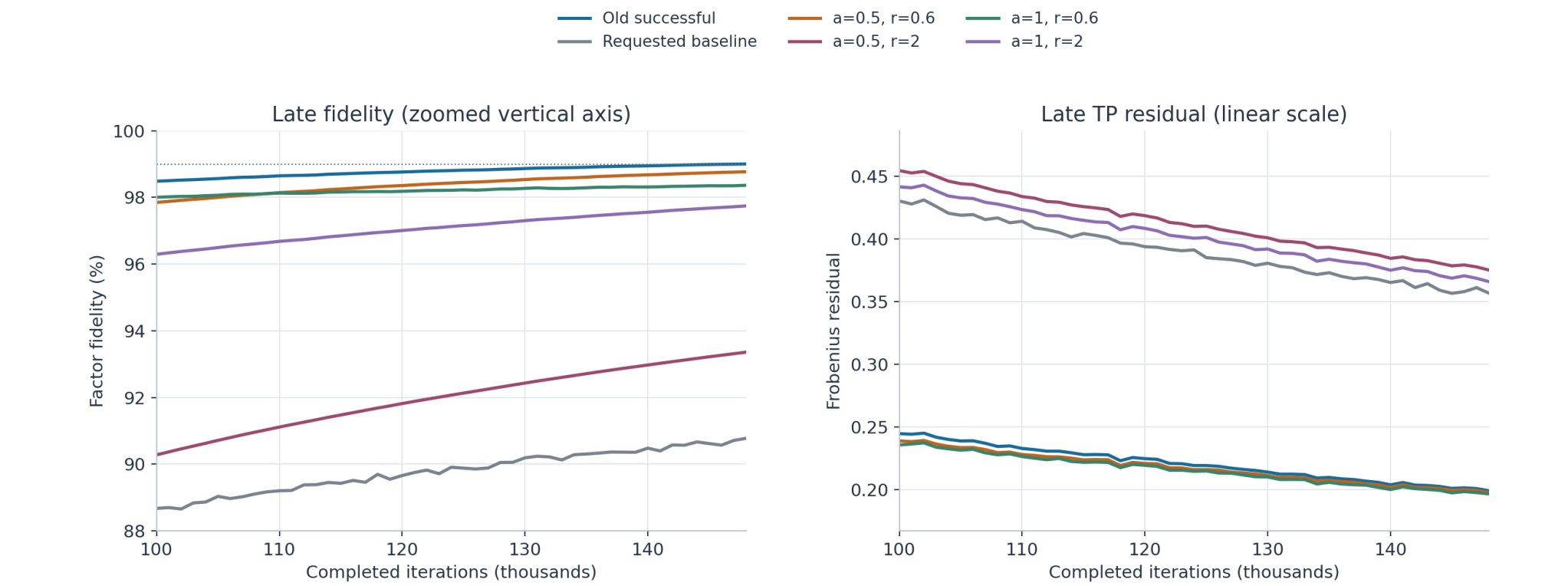


The best run is still improving

Recommendation: continue the best run unchanged from 148,000 to 200,000 total iterations. No job has been submitted.



Best case: a=0.5, r=0.6	100k-120k	120k-138k	138k-148k
Fidelity gain (percentage points)	0.5057	0.3001	0.1137
Gain per 10,000 updates (pp)	0.2528	0.1667	0.1137

Fidelity increased at every saved checkpoint after 100k, reaching 98.7689%; TP fell 17.2% over 100k-148k. Gains are slowing, but this trajectory is not flat. The scale effects interact: a=0.5 wins at r=0.6, while a=1 wins at r=2. No single scale change explains all four curves.

One targeted test, with the same exponents

Choice	Proposed setting / interpretation
Scientific question	Can more iterations alone close the remaining 0.2311 percentage-point gap to 99%?
Starting state	Exact restart of a=0.5, r=0.6 at step 148,000; retain factor, momentum, RNG state and absolute schedule index
Total / extra budget	200,000 total / 52,000 extra updates; 3,407,872,000 additional optimizer row draws
Step / smoothing / momentum	$\gamma=\min(1, 0.5/(k+10)^{3/4})$; $\beta=10^7/(k+1)^{1/4}$; $\rho=\min(1, 0.6/(k+4)^{1/2})$
Other settings	8 qubits; $\sigma=0.05$; B=65,536; rank 1; $\tau=20$; same data, fixed noise and validated optimized source
Resources / cost	One A100, 2 CPUs, 12G RAM, 30-minute cap; approximately 11.4 min runner / 6.9 min optimizer from measured scaling
Outcome / stopping	Run the full fixed budget. Verify endpoint $F\geq 99\%$; also report first crossing and whether all ten final checkpoints meet 99%. No automatic extension.

Why this budget is plausible, without treating extrapolation as a result

Descriptive fit to steps 100k-148k	Projected 99% crossing	Projected F at 200k
Linear fidelity	156.5k	99.814%
Exponential infidelity decay	163.8k	99.342%
Power-law infidelity decay	170.3k	99.205%

Earlier linear extrapolations overpredicted the held-out 148k fidelity by 0.57-1.71 pp. Exponential-error fits overpredicted by 0.15-0.30 pp; power fits had errors -0.047 and +0.014 pp in two retrospective checks. Recent-window power fits suggest roughly 170k-175k for a crossing. These are model-dependent scenarios, not confidence intervals; a fidelity plateau below 99% remains possible.

Changing β or other constants now would combine an iteration-budget test with retuning. This continuation keeps the schedule intact. Timing excludes queueing and separate setup/verification; CPU preparation overlaps GPU work. Report TP and sampled loss separately from the fidelity target.

Draft update for coauthors

Subject: 8-qubit QPT update with the paper's exponents

We completed four fresh 8-qubit runs with the usual Gaussian noise $\sigma=0.05$, batch 65,536 and 148,000 iterations, retaining step/smoothing/momentum exponents 3/4, 1/4, 1/2. Only the step and momentum scales changed; the dataset, initialization, row sequence and smoothing schedule were fixed.

The best pair was step scale 0.5 and momentum scale 0.6: 98.7689% factor fidelity and TP residual 0.19768, compared with 90.7749% and 0.35670 for the untuned requested-exponent baseline. The other scale pairs reached 93.3581%, 98.3634% and 97.7448%. None of the new runs reached 99%; the earlier schedule reached 99.0027% at the same iteration budget. These are one-channel, one-seed comparisons.

The best trajectory was still improving at every saved late checkpoint, although gains were slowing. Our proposed next test is to resume it unchanged to 200,000 total steps, preserving momentum, RNG state and schedule index. This is 52,000 additional updates on one A100, with an estimated runner time of about 12 minutes. It has not been submitted. Extrapolation suggests a crossing is plausible, but does not rule out a plateau below 99%.

We also removed redundant CPU row decoding/re-encoding while preserving exact sampled symbols and fixed noise. Host preparation improved 4.10x; three alternating pairs of 1,000-step trials gave 1.786x median runner speedup with bitwise-equal factors, momentum and diagnostics. Production host preparation fell from about 61.5 to 16.6-16.8 minutes; it overlaps GPU work. Validation used standalone exact checks; no pytest suite was run.

Fidelity here is $|c+u|^2/(|c|^2+|u|^2)$, not the measurement-loss term f , and it does not certify TP. The report includes solid fidelity/loss/TP curves, exact schedules, timing definitions and full provenance.

Records and scope

The full verified comparison remains in `qpt_n8_scale_tuning_20260923.pdf`. This addendum uses its saved scalar traces; it introduces no new QPT results. Analysis details and model definitions are in `followups_20260923/late_analysis.json`; exact continuation parameters, command arguments, source/data/restart hashes and validation criteria are in `proposed_experiment.json`. The editable message draft is `coauthor_summary.md`.

The proposal is unsubmitted and the message is unsent. Automatic campaign follow-up remains paused. Any later run must use a new guarded output directory, preserve the parent archive, and validate the resumed state and complete final factors within the allocation.